

CSE 3150 - C++ Essentials - Song Han - Spring 2025

Programming Assignment 8 – Due May 5th

1. Overview:

This assignment involves building a C++ program to simulate a digital library system. The system handles users, digital media (like eBooks, audiobooks, and online courses), rentals, and a media catalog. It showcases object-oriented programming techniques including class inheritance, polymorphism, the strategy pattern, and the singleton pattern.

2. Objectives:

- Apply object-oriented design principles effectively
- Implement software design patterns like Singleton and Strategy
- Simulate the core components of a digital media rental platform

3. Description of the Classes:

a. Media Catalog Class

- Purpose: Manages the digital media collection (inventory of eBooks, audiobooks, and courses)
- Functions: Add, remove, and check availability of media
- Implemented as a Singleton, ensuring a single catalog across the application

b. User Classes

- *User (Base Class)*
 - Attributes: User ID, name, subscription level, loyalty points, and payment method
 - Functions: Getters, access to payment method, and access to a personal RentalCart
- *StudentUser / FacultyUser (Derived Classes)*
 - Override calculateRentalDiscount() to offer different discounts:
 - StudentUser: 15% discount if loyalty points ≥ 100
 - FacultyUser: 25% discount if loyalty points ≥ 100
 - Subtract 100 loyalty points every time a discount is applied
 - Add loyalty points after purchase to respective user (loyalty $+=$ total / 10)

c. RentalCart Class

- Each user has an associated RentalCart
- Functions: Add media to cart, remove, clear cart, calculate total rental fee after discount

d. Payment Classes (Strategy Pattern)

- *PaymentMethod (Interface)*
 - processPayment(double amount)
- *CreditCard / PayPal (Concrete Strategies)*

- CreditCard allows overdraft up to \$50 (negative balance of -\$50)
- PayPal disallows overdraft (balance must be sufficient)

4. Steps of the System:

1. Media Creation and Catalog Management:

- Digital media items (eBooks, Audiobooks, and Online Courses) are created directly and added to the MediaCatalog (a Singleton), which maintains a centralized inventory
- The catalog supports adding, removing, and checking availability of media items

2. User Creation:

- Users are instantiated directly as StudentUser or FacultyUser objects, with attributes like user ID, name, subscription level, payment method and loyalty points
- Each user is assigned a RentalCart to manage their media rentals and payment method to purchase items

3. Media Rental Process:

- Users can browse and add available media to their RentalCart, remove media, or clear the cart entirely
- When ready to rent, the system calculates the total rental fee, applying discounts based on user type and loyalty points

4. Payment Handling:

- The user selects a payment method (either CreditCard or PayPal) to complete the transaction.
- The system uses the Strategy Pattern to process payment through the selected payment method:
 - CreditCard: Allows overdraft (i.e., payment can go through even if balance goes negative)
 - PayPal: Does not allow overdraft—payment fails if balance is insufficient
- If payment is successful, the transaction is completed, loyalty points are updated, and balance is changed.
- If payment fails, the transaction is canceled, points are not used or updated, and the balance stays the same.

5. Assignment Task:

- Implement the given classes according to the provided code snippets and descriptions
- Ensure the correct use of object-oriented principles and design patterns